MIL-A-52100B
21 September 1971
SUPERSEDING
MIL-A-52100A
3 June 1964

MILITARY SPECIFICATION

ADAPTERS: FOR BRIDGE, FLOATING, ALUMINUM, HIGHWAY-TYPE, DECK-BALK SUPERSTRUCTURE ON PNEUMATIC FLOATS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers adapters for use with the bridge, floating; aluminum; highway-type; deck-balk superstructure.

1.2 Classification. The adapters shall be of the following types, as specified:

Type I - Adapter, saddle, deck balk, normal.
Type II - Adapter, saddle, deck balk, offset.
Type III - Adapter, curb, deck balk.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

Federal

FED. STD. No. 595	- Colors.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

FSC 5420

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DRAWINGS

ME

D13218E9941	- Adapter, Saddle, Deck Balk, Steel, Normal.
D13218E9942	- Adapter, Saddle, Deck Balk, Steel, Offset.
DL13208E9900	- Adapter, Curb, Deck Balk.
D13217E7895	- Adapter, Saddle, Deck-Balk, Steel, Normal, Packaging of.
D13217E7896	- Adapter, Saddle, Deck-Balk, Steel, Offset, Packaging of.
D13217E7897	- Adapter, Curb, Deck-Balk, Packaging of.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY

A5.1 - Mild Steel Covered Arc-Welding Electrodes.
A5.5 - Low-Alloy Steel Covered Arc-Welding Electrodes.
B3.0 - Standard Qualification Procedure.
D2.0 - Welded Highway and Railway Bridges.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y. 10017.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street NW, Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Description. The adapters shall be in accordance with the drawings specified in 3.8 through 3.10, as applicable, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish an adapter of the type specified, for examination and testing, within the time frame specified to prove, prior to starting production, that his production methods will produce adapters that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings.

3.4 Welding electrodes. Welding electrodes for structural steel shall conform to AWS A5.1, E60XX series. Welding electrodes for low-alloy steel shall conform to AWS A5.5, E70XX series.

3.5 Identification marking. The adapters shall be metal-die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.6 Cleaning, treatment, and painting. All parts of the adapters shall be cleaned, treated, and painted in accordance with MIL-T-704, type B, except that the color number designation shall be 34087 in accordance with FED. STD. No. 595. Contact surfaces, regardless of whether such contact is between similar or dissimilar metals, shall be cleaned, treated, and painted prior to assembly.

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3.7 Stenciling. The nomenclature, Federal stock number, weight, and cube shall be stenciled on each type I and type II adapter. The letters "NET WT" shall precede the numerals in the weight and the letters "CU" shall precede the cubic displacement. Markings shall be Gothic-type capitals and Arabic numerals 3/4 inch high. Stenciling shall conform to MIL-T-704.

3.8 Type I adapter, normal. The type I adapter shall be as shown on Drawing D13218E9941 and as specified herein.

3.9 Type II adapter, offset. The type II adapter shall be as shown on Drawing D13218E9942 and as specified herein.

3.10 Type III adapter, curb. The type III adapter shall be as shown on DL13208E9900 and as specified herein.

3.11 Workmanship.

3.11.1 Steel fabrication. Steel used in the fabrication of the adapter shall be free from laminations, kinks, and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting with a tip suitable for the thickness of metal may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. All bends shall be made with metal dies or fixtures to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. All burrs, rough edges, and edges of holes shall be ground smooth.

3.11.2 Welders and welding.

3.11.2.1 Welders. Before assigning any welder or automatic welder equipment operator to welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder or operator has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society (B3.0).
Welding Qualifications of the ASME Boiler and Pressure Vessel Code (Section IX)

3.11.2.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the

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provisions of the American Welding Society publication "D2.0 - Welded Highway and Railway Bridges". All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire brushed clean.

3.11.2.3 Jigs and fixtures. Components, except for minor parts, shall be assembled in steel jigs or frames and welded while securely held in position. Minor parts may be held securely and tightly in position by any means that will insure secure and proper positioning. The jigs or frames shall be designed so that distortion of the parts will be held at a minimum.

3.11.3 Interchangeability. Like adapters shall be fabricated to be duplicates within the tolerances specified and shall be interchangeable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the supplementary quality assurance provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance or rejection (see 6.4 and 6.5).

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

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4.3 Preproduction inspection.

4.3.1 Examination. The preproduction adapter shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The adapters shall be examined for the following defects:

Major

101. Material not as specified.
102. Workmanship not as specified.
103. Incomplete or incorrect assembly.
104. Any part or component not in accordance with SQAP requirements as shown on the drawings.
105. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal.

Minor.

201. Dimensions other than those identified by SQAP data not as specified.
202. Identification marking incorrect, missing, or illegible.
203. Cleaning, treatment, and painting not as specified.
204. Stenciling not as specified for type I and type II adapters.

4.6 Inspection of preparation for delivery.

4.6.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2 Sampling. Sampling for examination shall be accordance with MIL-STD-105.

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4.6.3 Examination. Samples selected in accordance with 4.6.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106.	Type I adapters not nested and bundled together in the quantities required and in accordance with the referenced drawings.
107.	Type II adapters not nested and bundled together in the quantities required and in accordance with the referenced drawings.
108.	Type III adapters not packed in the quantities required and in accordance with the referenced drawings for level A or B.
109.	Marking illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified.

5.1.1 Level A.

5.1.1.1 Type I adapters. Five type I adapters shall be nested and bundled as shown in Drawing D13217E7895.

5.1.1.2 Type II adapters. Four type II adapters shall be nested and bundled as shown in Drawing D13217E7896.

5.1.1.3 Type III adapters. Eight type III adapters shall be packed in a close-fitting box as specified for level A as shown in Drawing D13217E7897.

5.1.2 Level B.

5.1.2.1 Type I and type II adapters. The type I and type II adapters shall be packed as specified for level A.

5.1.2.2 Type III adapters. Eight type III adapters shall be packed in a close-fitting box as specified for level B as shown in Drawing D13217E7897.

5.1.3 Level C.

5.1.3.1 Type I and type II adapters. The type I and type II adapters shall be packed as specified for level A.

5.1.3.2 Type III adapters. The type III adapters shall be packed to assure carrier acceptance and safe delivery to destination at lowest rating in a wooden container complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking. Shipping containers and bundles shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The saddle adapters, both normal and offset, are to be used as intermediate supports between aluminum deck-balk superstructures and 24-ton pneumatic floats. The curb adapter is to be used to raise a row of balk to provide curbing approximately 6 inches above the normal deck elevation.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of adapter required (see 1.2).
(c) Time frame required for submission of preproduction model (see 3.2).
(d) Level of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production adapters from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish adapters conforming to this specification.

6.4 Supplementary quality assurance provisions (SQAP). The contracting officer should require the supplier to maintain records of all SQAP inspections. A suggested paragraph is as follows:

"The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiences found, quantity approved, quantity rejected, corrected action taken when applicable."

6.5 Definitions.

6.5.1 Supplementary quality assurance provisions (SQAP). A SQAP is a contractual requirement that supplements section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

Custodian:

Army - ME

User activity:

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Preparing activity:

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Project No. 5420-0117

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION **MIL-A-52100B Adapters: For Bridge, Floating, Aluminum, Highway-Type, Deck-Balk Superstructure on Pneumatic Floats**

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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